

IRR shortcut methods

◆ 1. Meaning of IRR (Internal Rate of Return)

IRR is the **rate (r)** at which the **Net Present Value (NPV)** of all project cash flows equals **zero**.

$$NPV = 0 = \sum \frac{CF_t}{(1+r)^t} - \text{Initial Investment}$$

It tells us the project's expected rate of return.

◆ SHORT-HAND / TRIAL METHOD (also called Approximation Method)

This is a **quick, manual** way to find IRR without iteration tools like Excel.

Step 1:

Find NPV at two discount rates:

- One gives **positive NPV** (lower rate)
- One gives **negative NPV** (higher rate)

Let these be:

- r_1 = Lower rate
- r_2 = Higher rate

and corresponding NPVs are NPV_1 and NPV_2 .

Step 2: Apply IRR Formula

$$IRR = r_1 + \frac{NPV_1}{NPV_1 - NPV_2} \times (r_2 - r_1)$$

Example

Year	Cash Flow (\$)
0	-10,000 (Investment)
1	6,000
2	6,000

Now test two rates:

At 10% →

$$NPV = -10,000 + \frac{6,000}{1.1} + \frac{6,000}{1.1^2} = +413.22$$

(Positive)

At 15% →

$$NPV = -10,000 + \frac{6,000}{1.15} + \frac{6,000}{1.15^2} = -348.75$$

(Negative)

Now apply formula:

$$IRR = 10 + \frac{413.22}{413.22 - (-348.75)} \times (15 - 10)$$
$$IRR = 10 + \frac{413.22}{761.97} \times 5 = 10 + 2.711 = ** 12.711$$

✅ IRR ≈ 13.6%

♦ DIRECT METHOD (for Equal Annual Cash Flows)

When annual inflows are equal (annuity form), use this shortcut:

$$PV \text{ Factor} = \frac{\text{Initial Investment}}{\text{Annual Cash Inflow}}$$

Then **find the discount rate** from PV Annuity Table (or calculator) that gives that PV factor for given years.

Example

Initial Investment = \$40,000

Annual Cash Inflow = \$12,500

Life = 8 years

$$PV Factor = \frac{40,000}{12,500} = 3.2$$

Now look at **8-year PV annuity factor table**, find the rate giving 3.2.

From table:

- 15% → 4.487
- 20% → 3.837
- 25% → 3.570
- **30% → 3.107**

So IRR ≈ **between 25% and 30%**, closer to 28%.

✅ IRR ≈ 28%

⚡ Summary

Method	When to Use	Steps
Short-hand / Trial Method	Uneven cash flows	Compute NPV at two rates → Apply interpolation formula
Direct Method	Equal annual inflows	Divide investment by inflow → Find rate from PV annuity table